

VENDOR PERFORMANCE ANALYSIS

Exploratory Data Analysis & Strategic Insights Report

About This Report

This report presents a comprehensive vendor performance analysis, leveraging a dataset of 10,692 vendor-brand transaction records. The analysis examines purchasing behavior, sales performance, inventory efficiency, and profitability dynamics across the supplier network.

The primary objective is to provide decision-makers with data-driven insights that support strategic procurement planning, vendor relationship management, and inventory optimization. Findings are structured to address six core research questions, each supported by statistical evidence and culminating in actionable recommendations.

All monetary figures are reported in USD. Analytical methods include descriptive statistics, correlation analysis, outlier detection, and hypothesis testing. Data cleaning procedures were applied prior to analysis to ensure the integrity of reported findings.

Dimension	Detail
Dataset Size	10,692 records
Variables Analyzed	18 financial & operational metrics
Analytical Methods	Descriptive statistics, correlation analysis, hypothesis testing
Data Cleaning	Removed zero-sales, negative gross profit, and negative margin records

Executive Summary

The analysis reveals significant structural concentration and efficiency disparities within the vendor network. Six key findings warrant immediate management attention:

65.69% Top 10 Vendor Share	\$2.71M Unsold Inventory Capital	72% Bulk Purchase Cost Reduction	198 Brands Needing Promotion
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- Vendor concentration risk is high: the top 10 vendors account for 65.69% of total purchases, creating supply chain vulnerability.
- Bulk purchasing delivers a 72% unit cost advantage (\$10.78 vs. \$39.07 per unit for small orders), validating volume-based procurement strategies.
- \$2.71M in unsold inventory capital is tied up across slow-moving stock, eroding cash flow and storage efficiency.
- 198 brands demonstrate high profit margins but low sales volumes, representing an untapped revenue opportunity through targeted marketing.
- Low-performing vendors paradoxically maintain higher average profit margins (41.57%) versus top-performing vendors (31.18%), suggesting scale-profitability trade-offs.
- Hypothesis testing confirms statistically significant differences in profitability models between vendor performance tiers (null hypothesis rejected).

Section 1: Exploratory Data Analysis

1.1 Summary Statistics

The dataset encompasses 10,692 transaction records spanning 16 variables, including purchase pricing, sales volumes, excise taxes, freight costs, gross profit, profit margin, stock turnover, the sales-to-purchase ratio and many more. Key distributional observations are summarized below.

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

1.2 Outlier Analysis

Several variables exhibit extreme right-skew and high standard deviations, indicating the presence of outliers that reflect genuine operational heterogeneity rather than data error.

Purchase & Actual Price

Maximum values of \$5,681.81 and \$7,499.99 substantially exceed the means of \$24.39 and \$35.64, respectively, reflecting a premium product tier with limited volume.

Freight Cost

Range spans from \$0.09 to \$257,032.07, suggesting significant variation in shipment size, route complexity, and logistics provider efficiency.

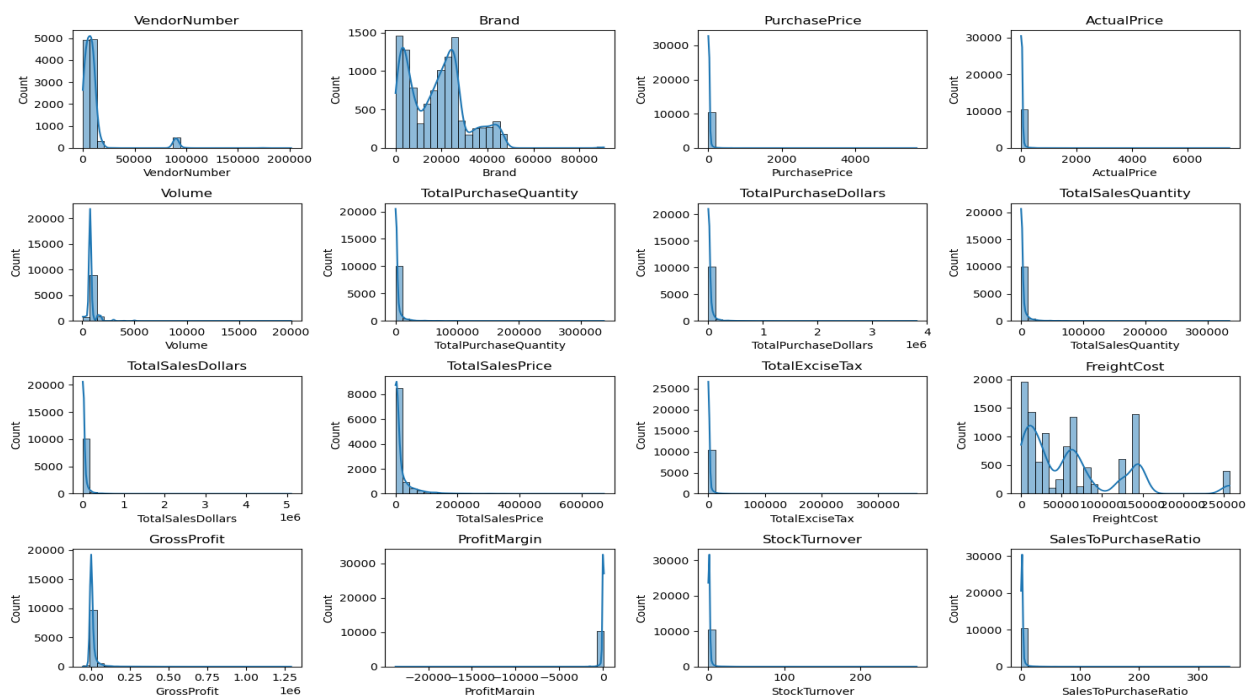
Stock Turnover

Values ranging from 0 to 274.5 indicate that certain SKUs fulfill backorders from legacy inventory, while others remain entirely unsold.

1.3 Anomalous & Zero-Value Records

Prior to analysis, records exhibiting the following characteristics were identified and excluded to ensure analytical integrity:

Condition	Implication	Action Taken
Gross Profit ≤ 0	Selling below cost or excessive discounting	Excluded from profitability analysis
Profit Margin ≤ 0 (incl. $-\infty$)	Zero or negative revenue scenarios	Excluded from margin benchmarking
Total Sales Quantity = 0	Purchased inventory never sold	Retained for inventory analysis; excluded from sales benchmarks

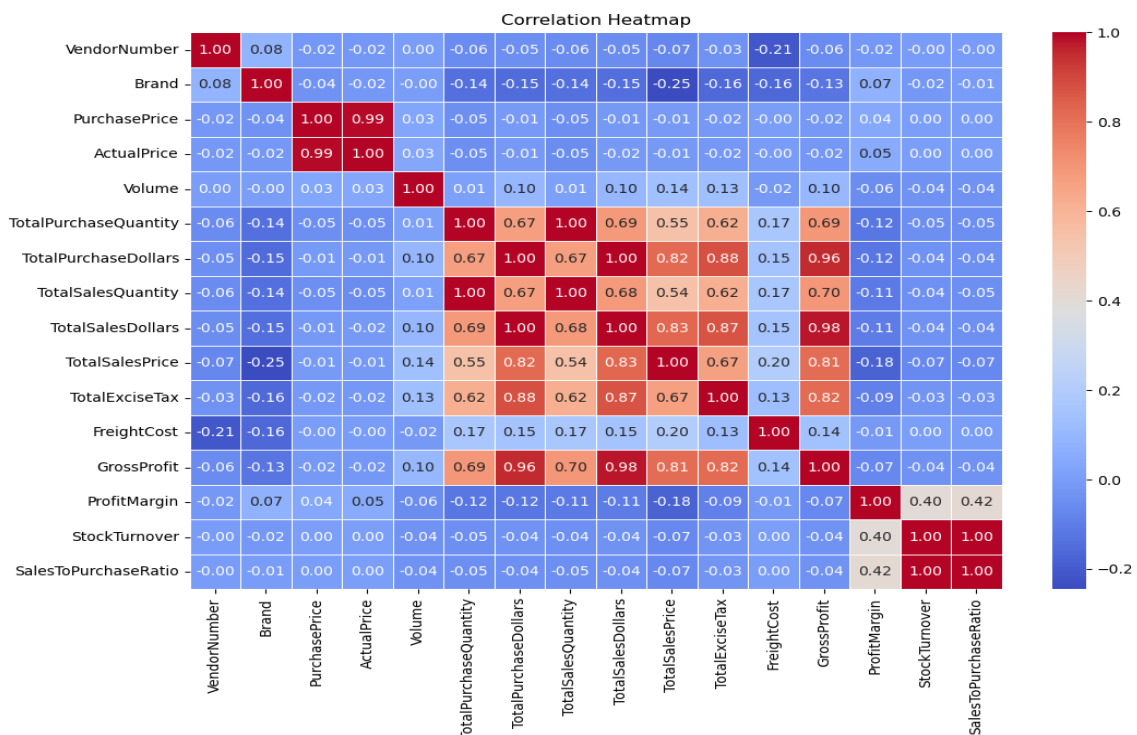


Section 2: Correlation Insights

A correlation heatmap was computed across 16 numerical variables. The following relationships are of primary strategic significance:

Variable Pair	Correlation Coefficient	Interpretation
Purchase Price vs. Total Sales Dollars	-0.01	Negligible; price variations do not drive revenue outcomes.
Purchase Price vs. Gross Profit	-0.01	Negligible; cost of goods has minimal direct profit impact.
Total Purchase Qty vs. Total Sales Qty	0.99	Near-perfect alignment confirms efficient inventory turnover.
Profit Margin vs. Total Sales Price	-0.17	Moderate negative; higher price points compress margins, likely due to competitive dynamics.
Stock Turnover vs. Gross Profit	-0.03	Weak negative; rapid turnover does not guarantee profit improvement.
Stock Turnover vs. Profit Margin	-0.05	Weak negative; volume velocity is decoupled from margin performance.

The near-perfect correlation between purchase and sales quantities ($r = 0.99$) confirms that the organization maintains an operationally efficient procurement-to-sales pipeline. However, the weak linkage between pricing and profitability signals that margin improvement must be pursued through cost optimization and strategic positioning rather than price increases alone.



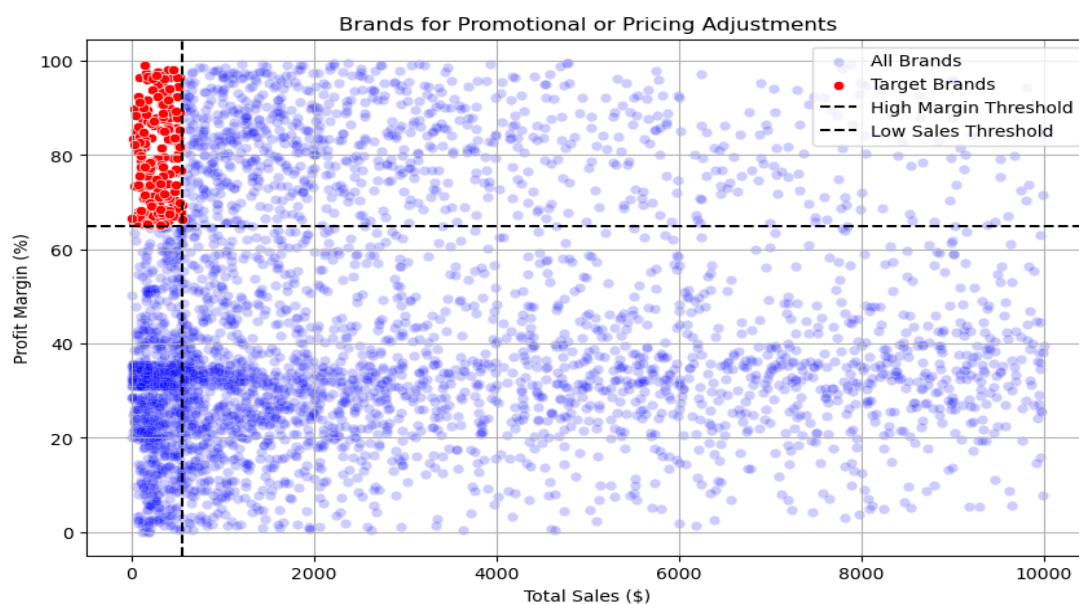
Section 3: Research Questions & Key Findings

Finding 1: Brands Requiring Promotional or Pricing Intervention

A total of 198 brands were identified as exhibiting low total sales dollars combined with above-threshold profit margins. These brands are positioned in a high-margin, low-volume quadrant, indicating that commercial potential is constrained by demand-side factors rather than structural cost issues.

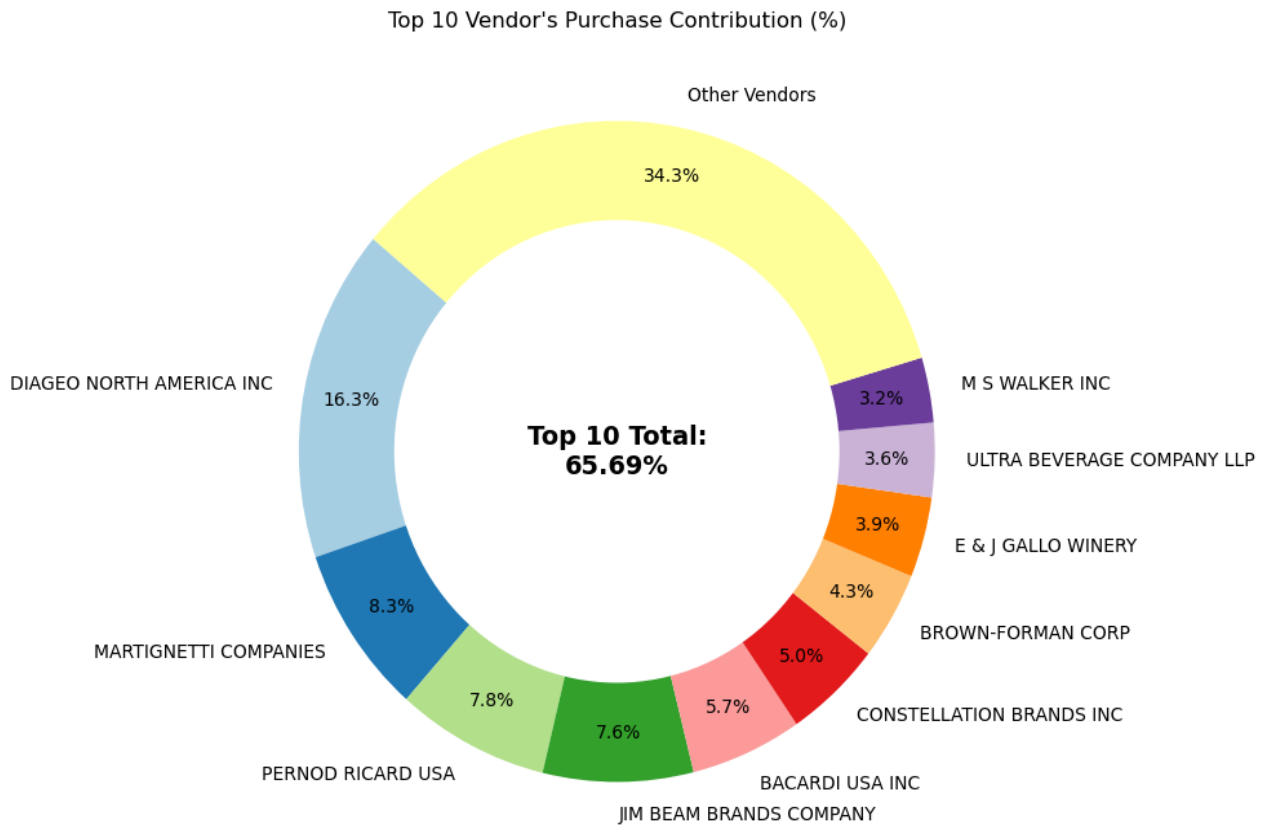
Brands with Low Sales but High Profit Margins:			
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752
198 rows × 3 columns			

Recommended interventions include targeted digital marketing campaigns, in-store promotional placements, and selective price reductions to stimulate trial and volume adoption without compromising margin architecture.



Finding 2: Vendor Concentration & Supply Chain Risk

Procurement is highly concentrated: the top 10 vendors collectively account for 65.69% of total purchase dollars, while the remaining vendor base contributes only 34.31%. This structural dependency introduces material supply chain risk.



A vendor diversification strategy is recommended to distribute procurement risk across a broader supplier base, particularly for high-volume SKU categories.

Finding 3: Bulk Purchasing & Unit Cost Efficiency

Order size analysis reveals a pronounced inverse relationship between purchase volume and unit cost, confirming economies of scale in procurement:

Order Size Category	Average Unit Purchase Price (\$)
Small	39.07
Medium	15.49
Large	10.78

Large-format orders achieve a 72% unit cost reduction relative to small orders. Organizations should evaluate inventory holding capacity and implement volume-tiered procurement agreements to systematically capture this efficiency.

Finding 4: Vendors with Low Inventory Turnover

Total unsold inventory capital stands at \$2.71M, distributed across vendors with stock turnover ratios below 1.0. Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

StockTurnover		VendorName UnsoldInventoryValue	
VendorName			
ALISA CARR BEVERAGES	0.615385	DIAGEO NORTH AMERICA INC	722.21K
HIGHLAND WINE MERCHANTS LLC	0.708333	JIM BEAM BRANDS COMPANY	554.67K
PARK STREET IMPORTS LLC	0.751306	PERNOD RICARD USA	470.63K
Circa Wines	0.755676	WILLIAM GRANT & SONS INC	401.96K
Dunn Wine Brokers	0.766022	E & J GALLO WINERY	228.28K
CENTEUR IMPORTS LLC	0.773953	SAZERAC CO INC	198.44K
SMOKY QUARTZ DISTILLERY LLC	0.783835	BROWN-FORMAN CORP	177.73K
TAMWORTH DISTILLING	0.797078	CONSTELLATION BRANDS INC	133.62K
THE IMPORTED GRAPE LLC	0.807569	MOET HENNESSY USA INC	126.48K
WALPOLE MTN VIEW WINERY	0.820548	REMY COINTREAU USA INC	118.60K

Recommended actions include renegotiating purchase quantities with low-turnover vendors, initiating clearance pricing on aged inventory, and establishing minimum turnover thresholds in vendor performance agreements.

Finding 5: Profit Margin Comparison — High vs. Low-Performing Vendors

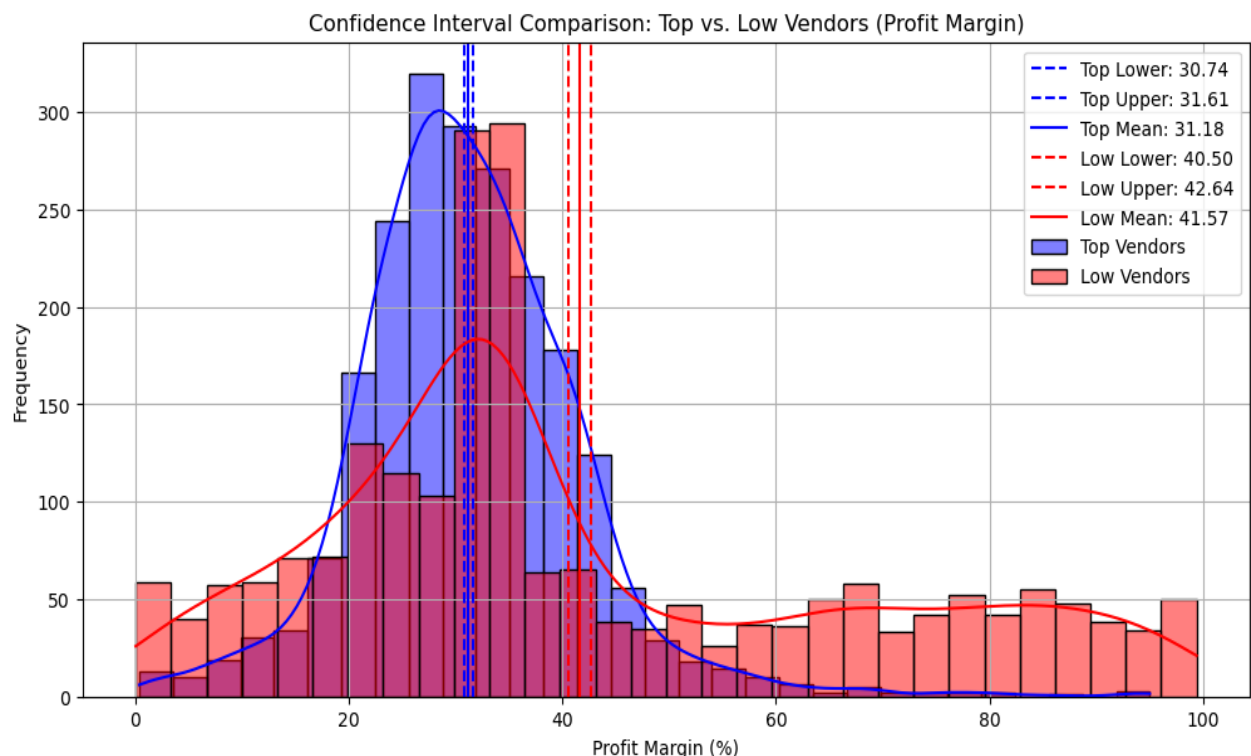
Counterintuitively, low-performing vendors (by sales volume) maintain statistically higher profit margins than top-performing vendors:

Vendor Tier	Mean Profit Margin (%)	95% Confidence Interval
Top-Performing Vendors	31.18%	(30.74%, 31.61%)
Low-Performing Vendors	41.57%	(40.50%, 42.64%)

This inversion suggests that low-performing vendors may be pricing at a premium that limits volume growth, while top-performing vendors leverage competitive pricing and distribution reach to maximize revenue at tighter margins. Both tiers require distinct optimization strategies.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies and enhance distribution networks.



Finding 6: Statistical Validation of Profitability Differentials

A formal hypothesis test was conducted to assess whether the profitability differential between vendor tiers is statistically significant:

Hypothesis	Statement
H ₀ (Null)	No significant difference in profit margins between top and low-performing vendors.
H ₁ (Alternative)	A statistically significant difference exists between the two vendor groups.
Test Result	Null hypothesis rejected. The two groups operate under structurally distinct profitability models.

This result confirms that vendor performance segmentation is analytically valid and that tailored strategies are warranted for each tier rather than a uniform approach.

Section 4: Strategic Recommendations

Priority	Recommendation	Expected Outcome
High	Re-evaluate pricing for 198 low-sales, high-margin brands through targeted promotions and demand stimulation campaigns.	Increased sales volume without sacrificing margin structure.
High	Diversify vendor partnerships to reduce the top-10 concentration from 65.69% to below 50% over 12 months.	Reduced supply chain disruption risk.
Medium	Formalize volume-tiered purchase agreements with key vendors to systematically capture the 72% bulk cost advantage.	Sustained unit cost efficiency and improved margins.
Medium	Address \$2.71M in slow-moving inventory through revised purchase quantities, clearance initiatives, and storage optimization.	Improved cash flow and reduced holding costs.
Medium	Deploy enhanced marketing and distribution strategies for low-performing vendors with high margin potential.	Margin-accretive volume growth for underperforming SKUs.
Low	Establish ongoing vendor performance scorecards incorporating turnover ratio, margin, and concentration metrics.	Data-driven vendor management and early risk identification.

Conclusion

This analysis provides a statistically grounded foundation for strategic vendor management decisions. The data reveals a vendor landscape characterized by significant concentration risk, notable profitability differentiation across performance tiers, and substantial untapped revenue potential in high-margin, low-volume brand segments.

The near-perfect alignment between purchase and sales quantities confirms operational efficiency at the logistics level; however, the \$2.71M unsold inventory capital and the 72% bulk cost differential signal that procurement and inventory strategies require structural refinement to fully realize the organization's profitability potential.

By implementing the recommendations outlined herein—particularly vendor diversification, targeted brand promotions, and bulk purchasing formalization—the organization is positioned to achieve sustainable revenue growth, reduced supply chain risk, and enhanced operational efficiency across the vendor network.

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